



UNIVERSITY *of* NICOSIA

Session

Prediction Markets

12 Week Schedule

We are here



Session	Session Name	Professor
1	Introduction to Blockchain and Web3	George Giaglis
2	Bitcoin and Digital Money	Garrick Hileman
3	Ethereum and Programmable Blockchains	Apostolos Kourtis
4	Digital Wallet Security and Privacy	Charles Guillemet (Ledger)
5	Decentralized Finance (DeFi)	Lambis Dionysopoulos
6	Prediction Markets	Apostolos Kourtis & Lambis Dionysopoulos
7	The Geopolitics of Cryptocurrency	Garrick Hileman
8	Tokenization and Stablecoins	Lambis Dionysopoulos & Lauren Berta (Ripple)
Advanced Track & MSc Students		
9	NFTs and Digital Ownership	Punk6529 & Mohsen El-Sayed (Ledger)
10	Regulation and Policy of Digital Assets	Jeff Bandman
11	Advanced Topics in Web3	Apostolos Kourtis
12	The Future: Convergence of Blockchain, AI, and IoT	George Giaglis

Session Objectives

By the end of this session, you will be able to:

- Understand what prediction markets are and how they work.
- Explain how market prices reflect probabilities of future events.
- Examine the role of blockchain infrastructure (smart contracts, oracles, traders).
- Compare prediction markets with polling and traditional betting markets.
- Discuss benefits and risks, including manipulation, insider trading, and regulation.

Agenda

1. Foundations and Mechanisms
2. Blockchain Implementation
3. The Prediction Market Landscape
4. Conclusions
5. Further Reading

Session: Prediction Markets

1. Foundations and Mechanisms

A multi-billion-dollar forecasting engine

- **Prediction markets** are markets where participants trade on the outcomes of future events
- Unlike traditional financial markets that price assets (equity, debt), prediction markets **price information**

The Wisdom of Crowds

The primary function of prediction markets is to aggregate vast amounts of disperse information. The market price acts as a real-time, continuously updated consensus forecast

Market Mechanics

Market Price	Implied Probability
Price Range	\$0.00 - \$1.00
Incentive	Truthful Revelation

Binary Payout Structure

Correct Outcome

\$1.00

Incorrect Outcome

\$0.00

The power of information aggregation

Information Aggregation	
Wisdom of Crowds	The aggregate judgment of a diverse group often exceeds that of a single expert
Incentive Alignment	Financial incentives motivate participants to research and reveal their private information and beliefs
Proven Accuracy	Success in forecasting political elections, economic indicators, and other measurable outcomes
Real-time Consensus	The market price acts as a real-time, continuously updated consensus forecast
Weighted by Conviction	Unlike polls, markets weight views by capital at risk, giving higher-conviction participants greater influence

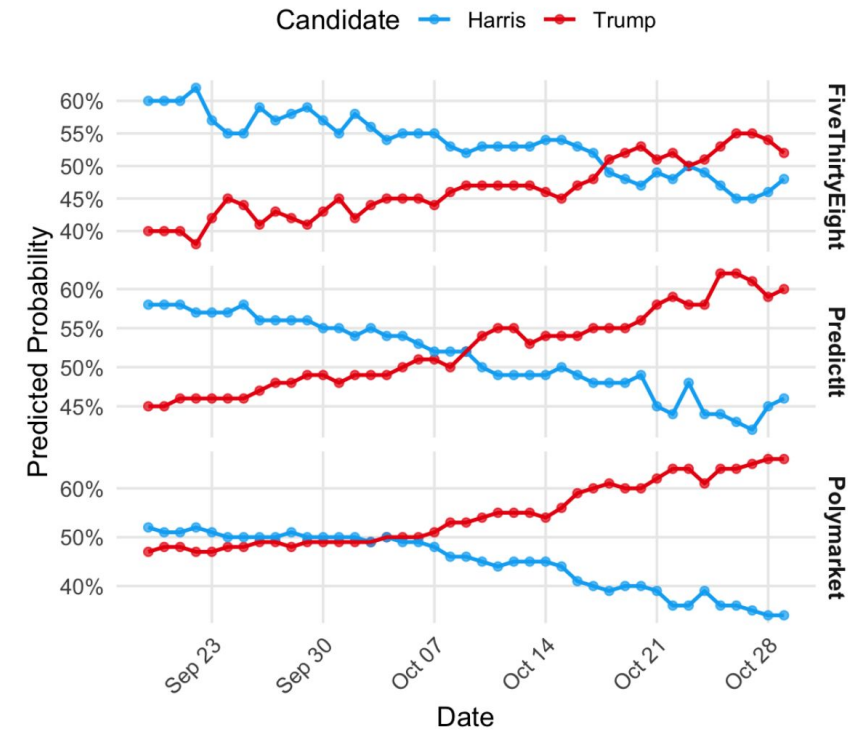
Prediction markets vs. polls

	Traditional Polling	Prediction Markets
Weighting	Equal / statistical	Capital-weighted
Incentive to be accurate	Low	High
Information aggregation	Snapshot survey	Continuous, real-time
Bias	Social desirability, sampling	Profit motive reduces noise
Probability output	Indirect (% support $\neq$ probability)	Direct (price = implied probability)

\$0.72 “YES” Share = 72% Implied Probability

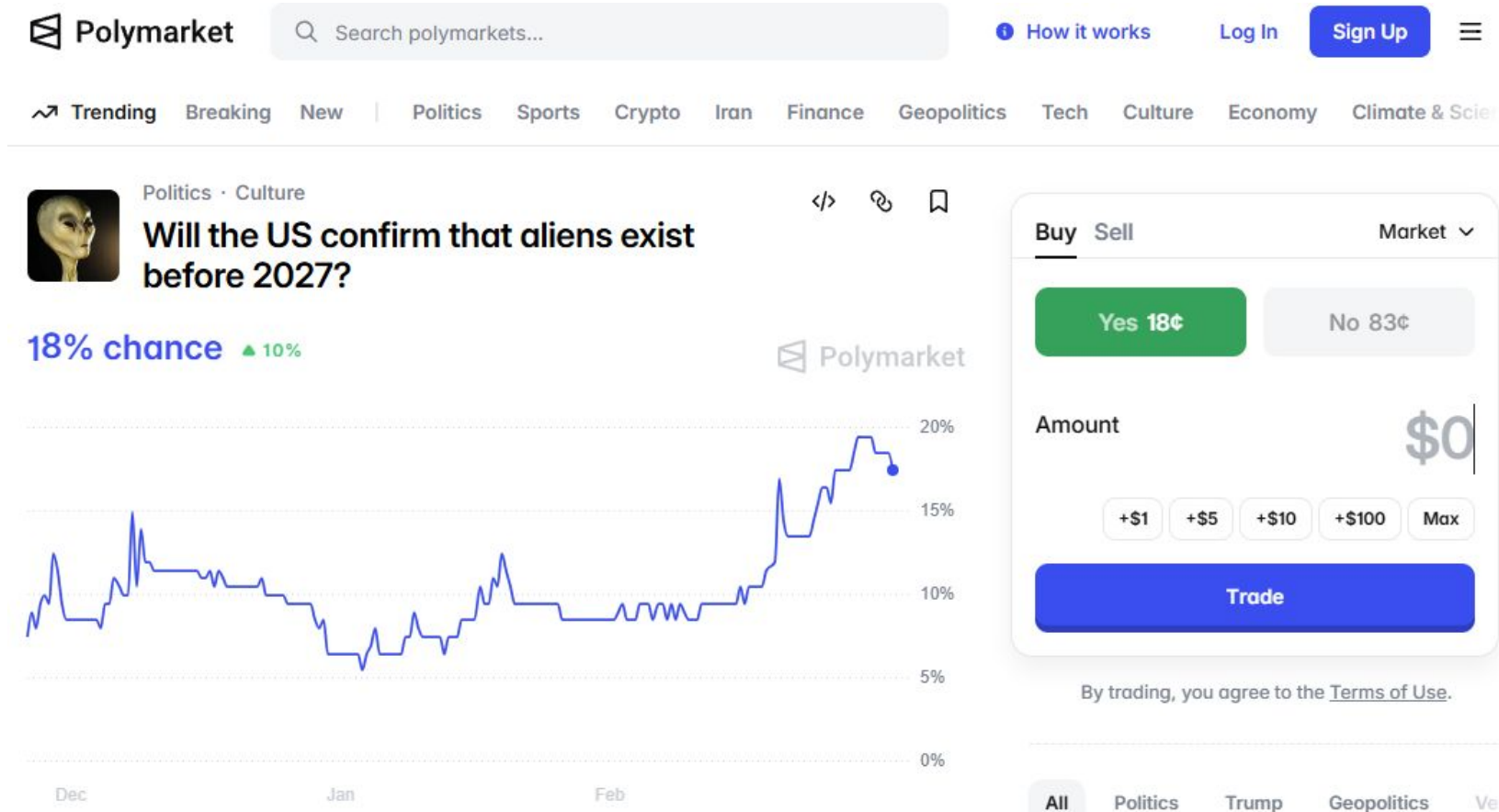
Why markets often outperform polls

- In prediction markets, investors put "their money where their mouth is" by placing **capital at risk**, unlike poll respondents who simply state preferences.
- Capital at risk can produce **more serious, better researched, and more informative signals** than polling data.
- Prediction markets have consistently outperformed polls in forecasting U.S. elections, with the Iowa Electronic Markets closer to the eventual outcome 74% of the time over 1988 to 2004 ([Berg et al., 2008](#), [Goodell et al., 2015](#)).
- There have also been instances of **insider trading**. This means that even more information is revealed compared to traditional markets.
- However, it is important to keep in mind that prediction markets can also be subject to biases and affected by **irrational investors** trading with their "heart" rather than "head".



Predicted probability for 2024 elections by market

An example



Source: [Polymarket](https://polymarket.com)

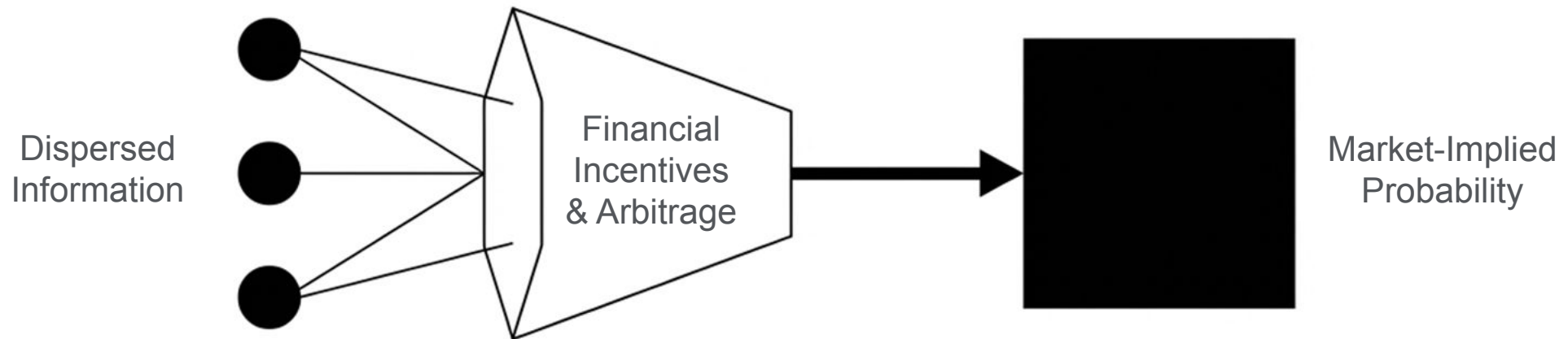
Reading prediction markets



Use cases

- **Forecasting events** such as elections, sporting events, and economic indicators
- **Risk hedging** or **speculating** related to shifts in interest rates, economic indicators, or broader market trends
- **Policy assessment**, i.e., assessing the outcomes and impacts of various policies or regulatory changes
- **Corporate decision-making** with businesses using internal prediction markets to forecast demand, project completion timelines, sales figures, and employee performance
- **Gathering distributed information** including rumours, private info, and local sentiment, which might not be captured by traditional polling methods

Economic foundations



Efficient Market Hypothesis (EMH)	Arbitrage	Information Discovery
EMH implies that asset prices reflect all available information. In prediction markets, the "asset" is a contract tied to a future outcome, and its price reflects the aggregated information and beliefs	Mispricing creates profit opportunities. Traders exploit these discrepancies, pushing the price toward the best estimate of the outcome probability	Financial incentives motivate participants to seek out new information and reveal it through trading

Federal (CFTC) vs. State regulation

A key question is whether prediction markets are financial derivatives under federal law or gambling regulated by the states.

Issue	CFTC View (Derivatives)	State Law View (Gambling)
Regulatory Authority	<u>The Commodity Futures Trading Commission (CFTC)</u> claims exclusive jurisdiction over event contracts under the Commodity Exchange Act (CEA)	States claim traditional authority over gambling regulation
Nature of the Activity	Platforms (e.g., Kalshi, Polymarket) argue contracts are financial derivatives used to trade expectations and generate price signals	Regulators argue contracts are bets on real-world events (sports, elections)
Regulatory Framework	Focus on market integrity, trading rules, financial risk	Focus on consumer protection, addiction controls, taxes, licensing
Legal Conflict	Federal framework may preempt state gambling laws for federally regulated exchanges	States argue prediction markets are backdoor sports betting circumventing state law

The history of prediction markets

Pre-1900s

Election betting markets operated in the **US** as early as the **1868 presidential race**. Election betting was common during **18th-19th century** parliamentary politics in the **UK**

1988

The **University of Iowa** established one of the first electronic academic prediction markets for U.S. elections. From 1988 to 2004, it outperformed polls about **74% of the time** ([Berg et al., 2008](#))

2000s

Ireland-based **Intrade** became the world's largest prediction market, attracting millions in volume. It faced a CFTC civil suit in 2012 and ceased trading in 2013

2018+

Augur (2018) pioneered decentralized prediction markets on Ethereum. **Polymarket** (2020) popularised on-chain prediction markets (\$20B+ annual volume, 2025). **Kalshi** launched publicly in 2021 as the first **federally regulated** U.S. exchange for event contracts

Session: Prediction Markets

2. Blockchain Implementation

The trust gap in centralized markets

Traditional prediction markets require **trust in a centralized provider**



Decentralization addresses critical limitations of centralized platforms

Counterparty Risk: You must trust the platform to hold your funds safely

Withdrawal Friction: Operators can freeze or delay payouts, or limit "winning" players

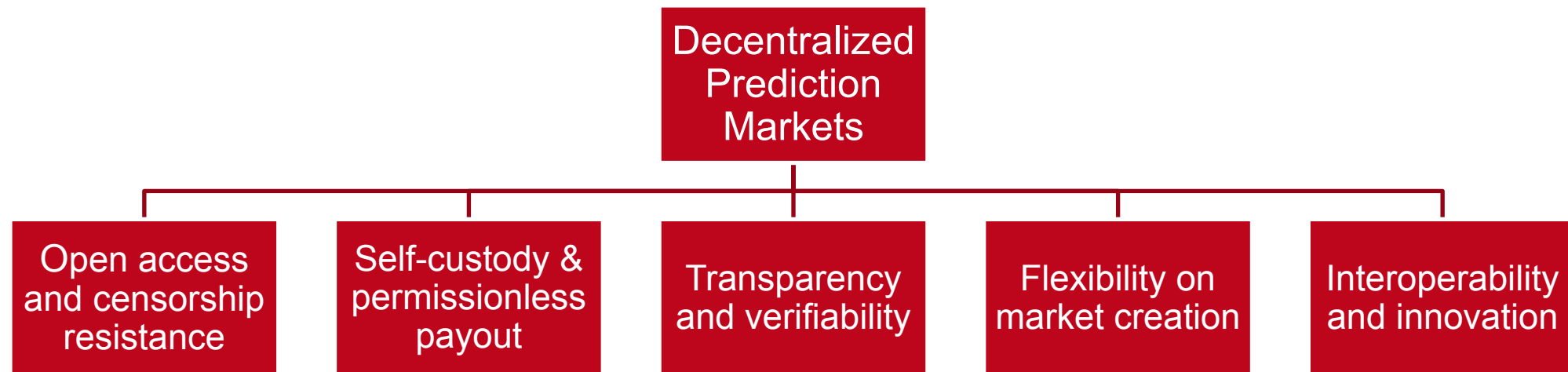
Opaque Settlement: The "house" decides the outcome

Geographic Barriers: Regulation can restrict access based on where you live

Market Listing Restrictions: Platforms decide which questions are allowed

Decentralized prediction markets

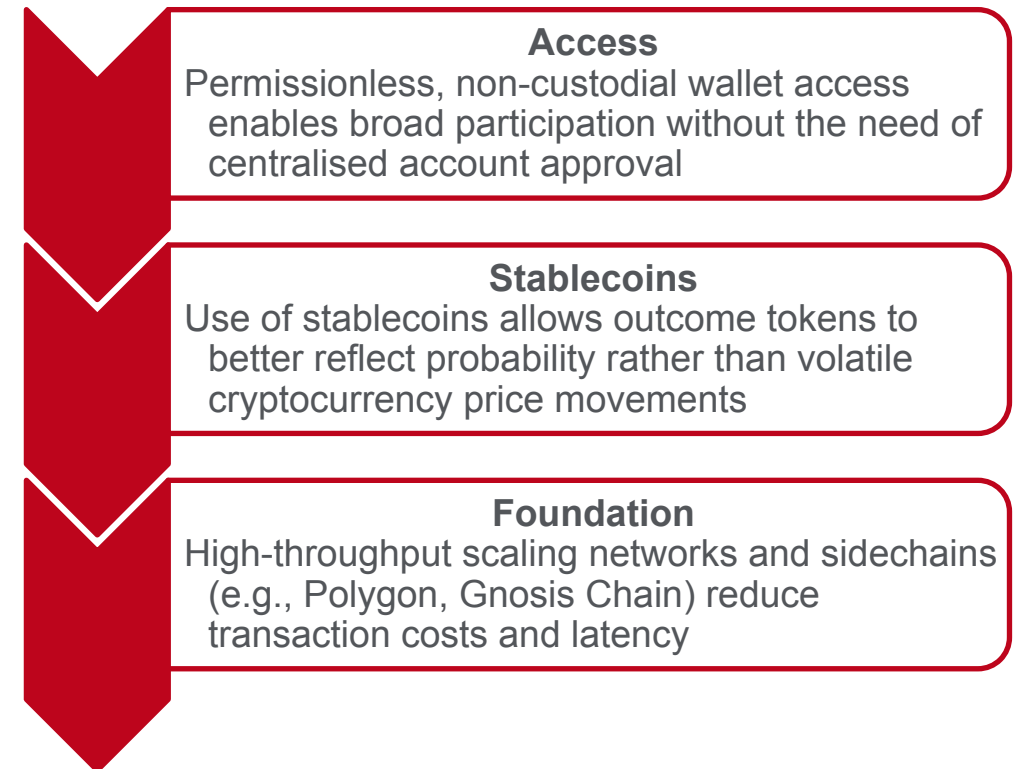
Decentralized prediction markets are blockchain-based markets that do not rely on a central operator, but instead use smart contracts and oracles to manage markets and settle results



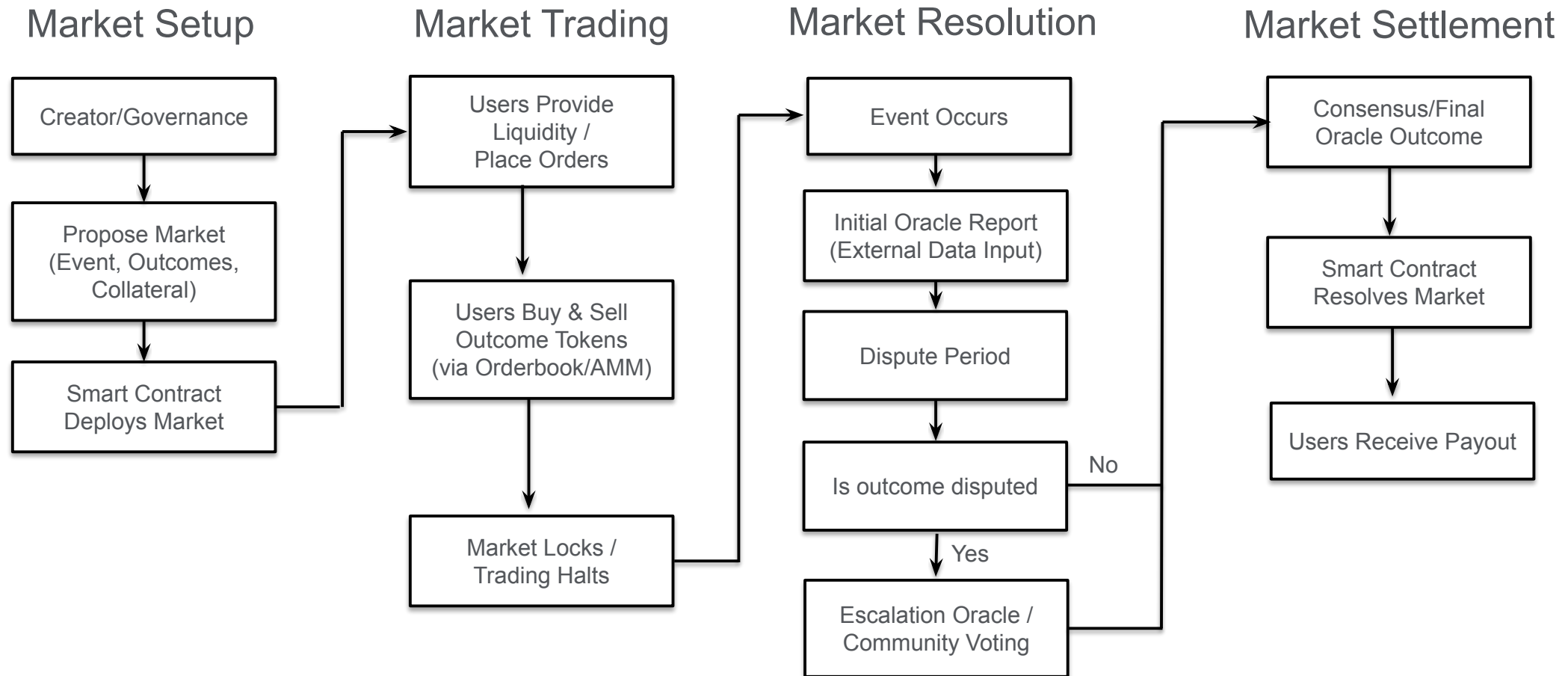
Key components and architecture

- **Smart contracts:** On-chain code that runs the market and holds funds
- **Oracles:** Systems that decide on and report the outcome
- **Liquidity providers:** Capital suppliers that support trading
- **Traders:** Participants whose transaction moves prices
- **Settlement & governance:** Mechanisms for resolving outcomes, paying winners, handling disputes, and updating the protocol

Decentralized architecture



The lifecycle of a decentralized prediction market



Smart contracts are the engine of trustless prediction markets

Smart contracts are self-executing programs that automate key functions of a prediction market, without the need of a central operator

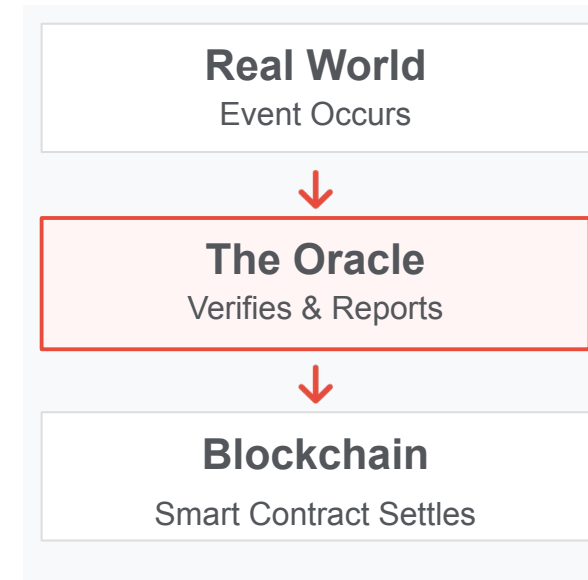
Core Functions
Create markets and define outcomes
Hold collateral securely
Hold collateral securely
Distribute payouts for settlement

Order Matching

	Continuous Double Auction (CDA)	Automated Market Maker (AMM)
Mechanism	Central limit order book where buyers and sellers submit limit/market orders; trades match continuously using price/time priority	Liquidity pool with algorithmic pricing (e.g., constant product ($x * y = k$)); traders swap directly against the pool
Used by	Polymarket, Kalshi (centralised) and others	Omen, Augur and others
Advantage	Highly efficient price discovery and tight spreads in liquid markets	Provides continuous liquidity as long as funds remain in the pool. Automated pricing for low-volume markets.
Disadvantage	Thin markets can have sparse order books and wide spreads	Low-liquidity pools can cause price slippage, and liquidity providers can suffer impermanent loss

Oracles

- The "**Oracle Problem**" is a fundamental challenge: blockchains cannot natively access real-world data
- **Oracles** serve as information bridges, feeding external information into smart contracts
- **Smart contracts** distribute funds based on the **outcome reported by the oracle**
- If the oracle is compromised, the entire market fails



Speed vs. Security

- **Centralized oracles** are fast but introduce trust assumptions
- **Decentralized oracles** are secure but slower and more expensive
- **Optimistic oracles** attempt to balance both

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3. The Prediction Market Landscape

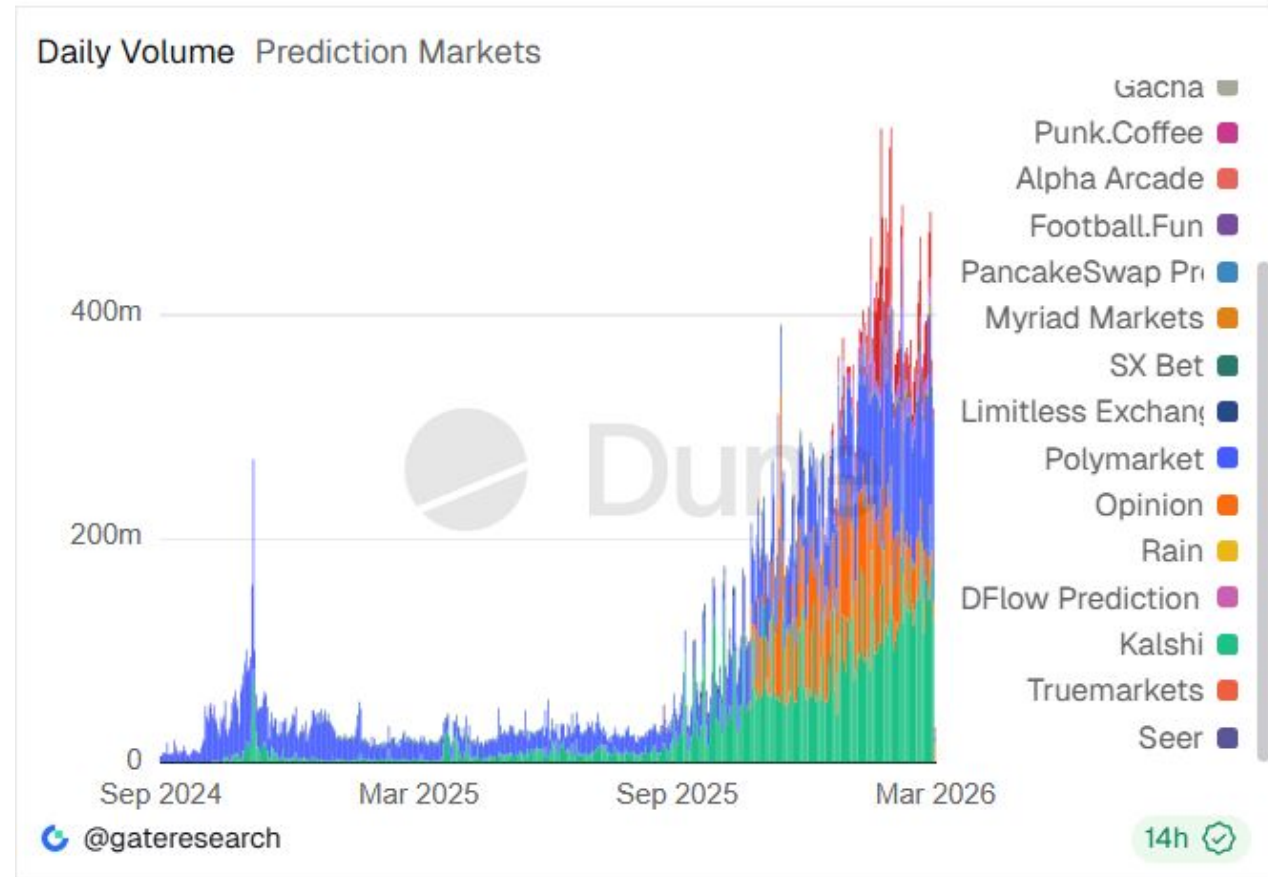
As we saw, prediction markets are growing fast

\$44B+

Total Industry Volume (2025)
Polymarket + Kalshi = ~88% share

600K+

Monthly Active Users
(industry peak 2025)



Source: [Dune Analytics](#)

Main competitors by volume

Name	TVL ↕	Volume 7d ↕	Fees 7d ↕	Revenue 7d ↕	Mcap/TVL ↕	Volume 30d ↕
1  Kalshi 1 chain		\$1.175b				\$4.788b
2  Polymarket 1 chain	\$386.91m	\$1.086b	\$1.63m	\$1.07m		\$4.057b
3  OPINION 1 chain	\$26.49m	\$81.4m	\$187,962	\$187,962		\$1.051b
4  Probable 1 chain	\$579,280	\$19.93m				\$1.034b
5  Predict Fun 2 chains	\$14.82m	\$31.52m	\$65,135	\$65,135		\$381.47m
6  Limitless Exchange 1 chain	\$503,920	\$28.22m	\$295,132	\$295,132	44.58	\$92.46m
7  SX Bet 1 chain	\$636,613	\$13.58m				\$70.42m
8  Myriad Markets 3 chains	\$688,057	\$29.13m	\$4,822	\$2,066		\$63.56m
9  PancakeSwap Prediction 1 chain		\$4.07m	\$122,075	\$122,075		\$14.82m
10  Football.Fun 1 chain	\$3.8m	\$706,781	\$21,721	\$21,721		\$4.63m

Source: [DeFiLlama](#)

Prediction markets and their cousins

Type	Regulation	Technology	Probability output	Example
Decentralized	Regulatory grey area, no formal oversight	Blockchain, smart contracts, oracles	Direct (price = implied probability)	Polymarket
Centralized	CFTC regulated (U.S.)	Traditional exchange infrastructure, fiat settlement	Direct (price = implied probability)	Kalshi
Poll aggregator	N/A (media product)	Bayesian and other simulation models	Model output (calibrated probability)	FiveThirtyEight
Traditional poll	Industry self regulation (e.g., AAPOR standards)	Survey methodology (phone, online, in person)	Indirect (% support is not a probability of winning)	Gallup
Betting exchange	Licensed gambling operator	Centralized matching engine, fiat settlement	Indirect (odds reflect bookmaker margin, not pure probability)	Betfair

Polymarket vs Kalshi

Category	Polymarket	Kalshi
Infrastructure	Decentralized (blockchain, smart contracts, oracles)	Centralised (traditionally exchange infrastructure)
Collateral	USDC (stablecoin)	USD (fiat)
Regulation	Regulatory grey area	CFTC regulated
Access	Permissionless, global, pseudonymous	KYC required, U.S. focused
Market creation	Open (bond required, ~\$750 USDC)	Platform controlled
Market resolution	Decentralized oracle with dispute mechanism	Platform administered

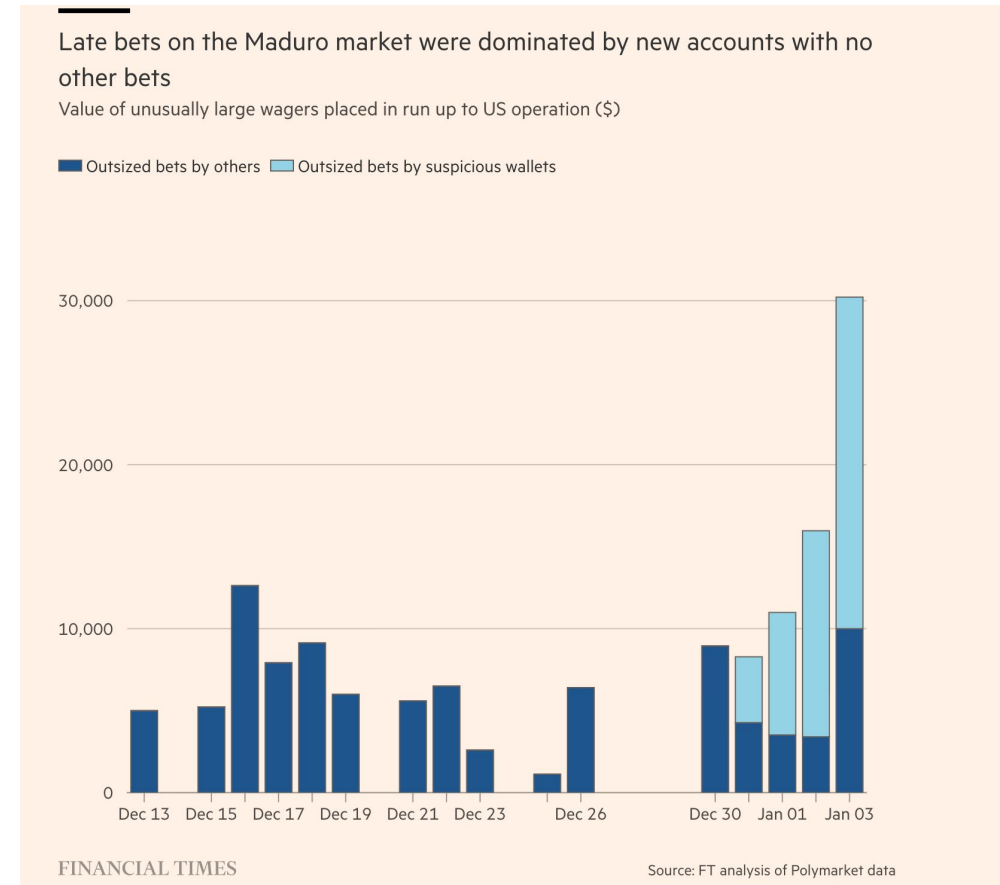
Insider trading in prediction markets

- People with **non-public information** (e.g., government officials or corporate insiders) can place bets before news becomes public and profit when market prices adjust.
- **Some argue informed trading improves predictions**, because insiders bring better information into the market.
- Others argue it **undermines fairness**, since ordinary participants cannot compete with traders who may already know the outcome.
- **Platforms differ in oversight:** e.g., Kalshi verifies users and monitors trades, while Polymarket allows anonymous crypto wallets with limited checks.
 - After a 2022 CFTC enforcement action, Polymarket was only recently allowed to return to the U.S. market; however, some users still access the platform via VPNs (which is not permitted), limiting oversight and enforcement.
- If users suspect insiders dominate trading, **trust in the market may decline**.

Sources: <https://www.ft.com/content/a46741cd-9bbc-4fd4-9a5d-9f860d36023c>, <https://www.ft.com/content/2883d3d4-aea2-4984-b994-4640593eed55>.

Insider trading example: Maduro capture

- When the US operation began on January 3, the odds of Maduro's removal were about **9%**.
- In early January, a Polymarket user made well-timed bets on the removal of Nicolás Maduro, reportedly earning **over \$400,000** when he was captured.
- In another instance, **six wallets** staked about \$9,800 and received roughly **\$133,000**.
- Several accounts **created shortly before the event** only bet on **Venezuela-related outcomes** and had **perfect records**, raising insider-trading concerns.



Source: [Financial Times](#)

Security risks, manipulation, and regulation

Manipulation risks:

- Pseudonymous trading enables wash trading and Sybil attacks
- Orders can be split across addresses, making whale activity harder to detect
- No formal mechanisms for addressing insider trading or market abuse
- Participant demographics may introduce systematic bias (e.g., crypto users skew right leaning per [Littrell et al., 2024](#))
- Information releases such as opinion polls can temporarily reduce price efficiency in prediction markets by attracting less experienced traders ([Brown et al., 2019](#))
- Ultimately, market prices reflect aggregated beliefs under specific market conditions, not objective likelihoods. ([Manski \(2006\)](#))

Technical risks:

- Oracle compromise is a single point of failure: incorrect outcome reporting means incorrect settlement
- In thin markets, the payoff from manipulating the oracle can exceed the cost of doing so
- Smart contract bugs can lead to loss of collateral (no recourse mechanism in fully decentralized systems)

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4. Conclusions

Conclusions

- Prediction markets aggregate dispersed information through financial incentives.
- Market prices provide real-time probability forecasts of future events.
- Blockchain enables decentralized markets, but introduces technical risks (e.g., oracle failures).
- Regulation and oversight vary across platforms, affecting transparency and trust.
- Prediction markets are powerful forecasting tools, but results reflect market conditions and participant beliefs, not perfect truth.

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5. Further Reading

Further Reading

Research Papers:

- [Distilling the wisdom of crowds: Prediction markets vs. prediction polls](#)
- [Prediction market accuracy in the long run](#)
- [When are prediction market prices most informative?](#)
- [How do financial markets price political uncertainty? Evidence from the 2024 United States presidential election](#)
- [Election uncertainty, economic policy uncertainty and financial market uncertainty: a prediction market analysis](#)
- [Political uncertainty and the 2012 US presidential election: A cointegration study of prediction markets, polls and a stand-out expert](#)



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